

Warm Up:

1. $(-5)(-6)$

→ 30

2. $-81 \div -9$

→ 9

3. $(-5)(-2)(3)$

→ $(10)(3) = 30$

4. $54 \div (-6)$

→ -9

5. A football team lost a total of 12 yards in two plays. What was the average loss per play?

→ $-12 \div 2 = -6$ yards per play

Speak Like A Mathematician:

variable: A letter used to represent an unknown or changing value.

algebraic expression: An expression containing at least one variable, along with numbers and operations.

term: A number, a variable, or a product/quotient of numbers and variables.

constant: A term that is a number only – its value never changes.

define a variable: To state clearly what the variable stands for in a situation.

Example 1: Write a Verbal Expression

- Write a verbal expression for $5x^3 + 2$.
→ five times a number cubed, increased by two
- Write a verbal expression for $(2/5)m^2$.
→ two-fifths of a number squared

Example 2: Write a Verbal Expression with Grouping

- Write a verbal expression for $a^4 + 6b/7$.
→ a number to the fourth power, plus the quotient of six times a second number and seven
- Write a verbal expression for $(a^4 + 6b) \div 7$.
→ the sum of a number to the fourth power and six times a second number, all divided by seven

Compare the two: the parentheses are the whole difference. In the second, the word "sum" must come BEFORE "divided by" to signal the grouping.

Example 3: Write an Algebraic Expression

- two times the quantity y plus 11
→ $2(y + 11)$ – "the quantity" signals grouping
- five less than a number cubed
→ $n^3 - 5$ (not $5 - n^3$)
- eighteen increased by the product of three and d
→ $18 + 3d$

Example 4: Write an Expression

- In the group play stage of the FIFA World Cup, teams are placed in groups of 4 and they play each other. A team is awarded 3 points for a win, 1 point for a tie, and no points for a loss. Write an algebraic expression that represents the number of points accumulated by one team in the group play stage.
→ $3w + 1t$, or simply $3w + t$, where w = number of wins and t = number of ties
- How would the expression change if a point were deducted for each loss?
→ $3w + t - \ell$, where ℓ = number of losses

Example 5: Evaluate an Algebraic Expression

- Evaluate $a^2(3b - a + 5) \div c$, if $a = 2$, $b = 6$, and $c = 4$.
→ $2^2(3 \cdot 6 - 2 + 5) \div 4 = 4(18 - 2 + 5) \div 4 = 4(21) \div 4 = 21$
- Evaluate $b(9 - c)/a^2$, if $a = 4$, $b = 6$, and $c = 8$.
→ $6(9 - 8) \div 4^2 = 6(1) \div 16 = 6/16 = 3/8$

Example 6: Write and Evaluate Expressions

- In 2004, Chad Fell set the record for the largest bubblegum bubble blown. Assume the bubble was spherical. The surface area of a sphere is four times π multiplied by the radius squared.

A) Write an expression that represents the surface area of a sphere.

$$\rightarrow 4\pi r^2$$

B) The record-setting bubble had a radius of 25.4 centimeters. Find the surface area of the bubble.

$$\rightarrow 4\pi(25.4)^2 = 4\pi(645.16) \approx 8107.3 \text{ square centimeters}$$

- The seating capacities of team stadiums in the AFC East division of the National Football League are shown below.

Team	Number of Seats
Miami Dolphins	65,326
New England Patriots	66,829
Buffalo Bills	71,608
New York Jets	82,500

A) Write an algebraic expression that represents the maximum number of attendees at the Jets, Patriots, and Dolphins home games during the season. Let j , p , and d be the number of home games for each.

$$\rightarrow 82,500j + 66,829p + 65,326d$$

B) After the sixth week, the Jets had played 4 home games, the Patriots 3, and the Dolphins 2. Find the maximum number of attendees.

$$\rightarrow 82,500(4) + 66,829(3) + 65,326(2) = 330,000 + 200,487 + 130,652 = 661,139 \text{ attendees}$$

Tonight's Homework: Paper Practice 1.2 (#1–12)